Statistical problems - statistical summaries

Task A: Problems with statistical measures based on the data

1) The four sets of data all have a mean of seven.
In which sets of data do you think that this average provides a good summary of the information? Give reasons for your answers.

a) 1, 1, 1, 1, 1, 1, 43 ✗ *43 is much bigger than all of the other numbers, which are all equal to 1.*

b) 7, 7, 7, 7, 7, 7, 7 ✓ *All of the numbers are 7.*

c) 1.5, 2.8, 6.1, 6.9, 7.9, 9.4, 14.4 ✓ *7 is around the centre of the data and many of the numbers are close to 7.*

d) 4, 5, 6, 7, 8, 9, 10 ✓ *7 is in the centre of the data. Each number that is below 7 has a matching pair which is just as much above 7.*

2) Find the mean, median and mode for the set of data.

1, 2, 3, 3, 4, 4, 5, 6, 9, 9, 9

mean = 5
median = 4
mode = 9

Which average least represents the central tendency for this data? Explain why.

the mode

The mode happens to be equal to the highest value in this data.

Task B: Choosing statistical measures based on the context

1) The table contains data from The Met Office about the weather in Cambridge in 2004. It shows the number of days in each month where there was frost.

month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
no. days	7	11	3	0	0	0	0	0	0	0	3	9

a) Find the mean number of days of frost in a month, giving your answer to 2 decimal places. *2.75*

b) Find the median number of days of frost in a month. *0*

c) Find the mode. *0*

d) Which measure of average would be the least useful for describing the amount of frost that could be expected during a typical month in Cambridge? Explain why.

the mean

It would suggest that each month typically has 2-3 days of frost, but most months do not have any frost.

2) A homeowner is selling their house. The data below shows the prices for the last five houses that were sold on the same street.

£150 000 £350 000 £360 000 £350 000 £210 000

a) Find the mean and median house price.

mean = £284 000
median = £350 000

b) Which average suggests that houses here are worth more money? *the median*

c) Which average suggests that houses here are worth less money? *the mean*

d) Which average might the seller prefer to use? *the median*

e) Which average might the buyer prefer to use? *the mean*